

Performance analysis of a hybrid aircraft propulsion system using solid oxide fuel cell, lithium ion battery and gas turbine

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HIGHLIGHTS

- A novel integration of SOFC, lithium ion batteries and gas turbine propulsion is proposed for hybrid electric aircraft.
- Compatibility and applicability of the proposed system are assessed.
- An afterburner generates high temperature streams for the gas turbine and SOFC reactant preheating.
- A water-cooled thermal management system is introduced to cool lithium ion batteries.
- Effects are investigated of key operating parameters on the hybrid system performance.

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ABSTRACT

A novel integration of a solid oxide fuel cell (SOFC), lithium ion batteries and a gas turbine propulsion system is proposed for a hybrid electric aircraft. Methane as hydrogen storage medium is fed to the propulsion system so as to avoid the limitations associated with the use of pure hydrogen storage and utilization. An internal reformer is used in the SOFC to produce hydrogen through the steam reforming and water–gas-shift reactions of the methane and steam mixture. An afterburner is used to generate the heat required for generation of electrical power in the gas turbine and preheating the SOFC reactants by burning the remaining fuel from the SOFC. Also, a pathway for liquid water input to the hybrid system is designed in a way to first circulate through the lithium ion batteries to remove generated heat from them before it is used in the remainder of system. The variations of heat generation rate and temperature of the battery with time are predicted through electrochemical and thermal models. A performance assessment of the proposed propulsion system is conducted to investigate the compatibility and applicability of the system. It is found that at 1C (charge/discharge rate of battery), the average temperature of the lithium ion battery is maintained at about 28 °C through a water-based cooling process. The optimum net electrical power output and energy efficiency of the hybrid propulsion system are achieved at compressor pressurization ratio of 7.5. Furthermore, the net electrical power produced and energy efficiency of the hybrid propulsion system reach their maximum values (i.e., 6.6×10^6 W and 46.5 %, respectively) at a SOFC temperature of 775 °C.

1. Introduction

Energy storage is becoming increasingly important. The annual increases in energy storage capacity by country are shown in Fig. 1 for the years 2015 to 2020. It is seen that in 2020, 5 GW was added to the global battery storage capacity, and that by the end of 2020, the total installed global battery power storage capacity was about 17 GW [1]. The total installed battery storage capacity grew by about 64 % in 2020 compared to 2017. China and the United States are leading in gigawatt-scale

addition of battery storage capacity. Note that about two-third of the total added capacity in Fig. 1 is associated with utility-scale installations [1].

Reducing carbon dioxide emissions from marine transportation though diesel-based propulsion systems of ships requires strict regulations [2]. Numerous projects have aimed at replacing diesel-based propulsion with fuel cell-battery combined systems. For instance, in 2017–2021, in the ReverCell project, proton exchange membrane (PEM) fuel cells with a total power output of 250 kW were combined with battery systems to drive hybrid ferries (Zero emissions, 2020). In

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Nomenclature			
A	cell area (m ²)	U_{O_2}	air utilization factor
a_i	activity factor of species i in the electrochemical reaction	U_{H_2}	hydrogen utilization factor
c_p	specific heat at constant pressure (J/kg.K)	V	volume (m ³)
c	salt concentration (mol/dm ³)	$\dot{W}$	work rate (W)
D	diffusion coefficient (m ² /s)	x	v
E	electrical potential (V)	x_i	molar concentration of species i
E_{OCF}	open circuit potential (V)	x_{sc}	steam to carbon ratio
F	Faraday constant (C/mol)	y	stoichiometric coefficient
$\bar{g}$	molar Gibbs free energy (kJ/kmol)	Z	altitude (m)
g	gravitational acceleration (m/s ²)	<i>Greek letters</i>	
$\bar{h}$	molar enthalpy (J/mol)	ϵ	volume fraction of polymer
i	current density (A/m ²)	ρ_B	density of battery (kg/m ³)
i_0	exchange current density (A/m ²)	ρ_{int}	electrical resistivity of interconnect (Ω .cm)
i_L	limiting current density (A/cm ²)	ρ_c	electrical resistivity of cathode (Ω .cm)
I	current (A)	ρ_e	electrical resistivity of electrolyte (Ω .cm)
J_n	pore wall flux of lithium ions (mol/(cm ² .s))	ρ_a	electrical resistivity of anode (Ω .cm)
k	constants for the reactions occurring at electrode	λ	effective water content of the membrane
K_s	equilibrium constant for the reaction	α	transfer coefficient
k_r	thermal conductivity in radial direction (W/m.K)	δ	thickness (m)
k_z	thermal conductivity in axial direction (W/m.K)	σ	electronic conductivity (S/cm)
L_a	base temperature lapse rate per kilometer of geopotential altitude (K/km)	$\gamma_{\pm}$	activity coefficient
$\overline{LHV}$	lower heating value (kJ/kmol)	γ_M	electrode surface roughness factor
m	mass (kg)	γ	Bruggeman coefficient
M	Mach number	γ_a	specific heat ratio of air
n_e	number of electrons in the balanced electrochemical half reaction	$\bar{A}_{s,0}$	ionic conductivity of electrolyte (S/cm)
N	number of cells	$\bar{A}_s^{eff}$	effective electrolyte ionic conductivity (S/cm)
$\dot{N}$	molar flow rate (mol/s)	∇t_+^0	transference number
P	pressure (Pa)	Φ	electrochemical potential (V)
P_i	partial pressure of component i (Pa)	η	overpotential (V)
$\dot{Q}$	heat transfer rate (W)	$\eta_{isen,T}$	isentropic efficiency of the gas turbine
$\dot{Q}_{gen}$	heat generation rate per unit volume for the battery (W/m ³)	<i>Subscripts</i>	
r	molar conversion rate	a	anode, activation, air
R	universal gas constant (J/(mol.K))	B	battery
R_a	characteristic gas constant for air (J/(mol.K))	c	cathode, concentration
R_s	particle radius (m)	SOFC	solid oxide fuel cell
$\bar{s}$	molar entropy (kJ/kmol.K)	ohm	ohmic
ΔS	entropy change due to electrochemical reactions (J/K)	ref	reference
t	time (s)	s	surface
T	temperature (K)	+	positive electrode
		-	negative electrode
		1	solid phase of electrode
		2	solution phase of electrode

addition, in 2021 a new project was initiated in which solid oxide fuel cells (SOFCs) with a total power output of 60 kW are integrated with a battery and internal combustion engine system to drive cruise ships [3]. Although fuel cell-battery based power systems have been successfully implemented in marine projects, several challenges like high manufacturing costs, fuel storage challenges, maintenance and safety issues need to be addressed to stimulate wider use of fuel cell-battery technology in ships.

The development of electric aircraft has recently received increasing global interest to reduce cost per passenger and greenhouse gas (GHG) emissions [4]. Several companies like Zunum Aero, Airbus and Eviation are investigating battery-based short range passenger aircraft. The hybridization of alternating propulsion systems and associated flight performance is addressed by Ji et al. [5]. Also, the National Aeronautics and Space Administration (NASA) has invested notably in electric and hybrid electric aircraft research, and expects larger prototypes to be test-ready by 2025 [6]. Collins and McLarty [4] proposed a fuel cell-gas

turbine hybrid system for an aircraft that utilizes liquid hydrogen as fuel. They describe the geometry and material requirements for a fuel cell-gas turbine-battery hybrid system. The performance of the proposed system is analyzed for four commercial aircraft flight profiles, and optimized for maximum payload capacity (i.e., the maximum weight it can safely carry). The authors show that the high energy density of liquid hydrogen and the high efficiency of solid oxide fuel cells decreases the fuel requirement by 70 %, which offsets the weight of the onboard electric power unit. Furthermore, Seyam et al. [7] propose a hybrid propulsion system for an aircraft in which a turbofan is integrated with a SOFC. Thermodynamic and parametric studies are carried out and the performance of the system is assessed through energy and exergy methods. Also, the use of various types of fuels such as methane, hydrogen, methanol and ethanol in numerous combinations is investigated as a means to replace kerosene, the conventional fossil-based fuel used in aircraft. The authors show that the integration of a SOFC with a conventional turbofan can improve system performance. Also, the use of

a fuel mixture of 40 % hydrogen and 60 % ethanol increases the performance by 5 % and decreases carbon emissions by 73 %.

Much research has been reported on electric and hybrid aviation vehicles. For instance, actual European flight information have been collected by Kousoulidou and Lonza [8] to compare the consumption of bio-kerosene and conventional kerosene in terms of carbon dioxide emissions. It was found that consuming 170 million tonnes of conventional kerosene results in emission of 400 million tonnes of GHG emissions by 2030. So, one European Union plan calls for the use of biofuels like hydrotreated vegetable oils (HVOs) and biomass-to-liquid (BTL) biojet fuel to reduce carbon dioxide emissions from the aviation transportation [8].

Wheeler et al. (2021) describe electric and hybrid electric aircraft propulsion systems, along with technological information and recent developments and research. Also, Sziroczak et al. (2020) discuss the technology, challenges, barriers and future directions for the hybrid electric aircrafts. Hoelzen et al. [9] emphasize the proper selection of battery systems in electric and hybrid electric aircraft, as the feasibility of such aircraft relies greatly on the specific energy of the battery. The authors demonstrate that the proper selection of the power-to-energy ratio of a battery based on the flight mission determines the technological feasibility of electric and hybrid electric aircraft. In addition, the German Aerospace Center (DLR) assessed various hybrid electric aircraft configurations for design mission of 800 nautical miles with 70 passengers. They study designs for various configurations of hybrid electric aircraft for entry into service in the year 2035 [10].

Ji et al. [11] propose a novel SOFC jet hybrid engine in which the SOFC is used instead of a turbine to generate electrical power for the compressor. They assess this hybrid engine thermodynamically and investigate the effect of flight altitude and pressure ratio of the hybrid engine on system performance. It is found that the thermal efficiency of the hybrid engine increases to 57.6 % and the highest exergy destruction rate is attributable to the SOFC and combustor.

Li et al. [12] design a fuel cell and battery hybrid propulsion system for light electric aircraft. The results of their numerical analysis indicate that the developed system can meet the electrical power requirement of the aircraft in cases of fast-charging and peak electrical power requirement. They show that, by using the battery energy storage system, the consumption of hydrogen can be reduced by 7.6 % compared to the use of a sole fuel cell in the electric aircraft.

Affonso et al. [13] compare conventional and hybrid electric propulsion systems for regional aircraft using energy and exergy methods. They show that, in conventional propulsion systems, more than 90 % of irreversibilities are associated with the thermal engine, and that hybridization of a propulsion system can reduce system irreversibilities,

while the integration of battery storage units can reduce further the exergy destruction in the system. They suggest higher degrees of hybridization for aircraft in the near future.

Guo et al. [14] combine a turbofan engine, solid oxide fuel cell and Al-H₂O reactor system for electrical power generation in unmanned aerial vehicles. The Al-H₂O hydrogen reactor generates hydrogen helps reduce the specific fuel consumption and meet the increasing power demand in the unmanned aerial vehicle. They develop a mathematical model for the components of the propulsion system and show that the combined system can reduce fuel consumption by more than 10 % compared to regular turbofan engines.

In mobile applications of battery systems in the marine, aviation and road transportation vehicles, the thermal management of battery cells is one of the key factors to be considered in vehicle design. The battery thermal management system maintains the battery temperature within the desired operating range normally 15 °C–45 °C). Although battery thermal management systems are usually designed to cool battery systems, they should be able to provide heat for low environmental temperatures and initial operation conditions of batteries. Since exceeding battery temperature from desired operating range may result in thermal runaway phenomenon (i.e., allowing the battery temperature to exceed the critical point that may lead to fire and explosion), cooling of battery cells is typically the main concern in the battery thermal management research. There are three main types of battery thermal management systems: air-based [15,16,41], liquid-based [17,18] and phase change material-based. The latter category of battery thermal management systems includes systems based on solid–liquid phase change [19,20,21,22,23] and liquid–vapor phase change [24,25].

The use of fuel cells and batteries for electricity production and storage in electric aircraft propulsion systems can help in achieving or shifting towards net-zero carbon emissions. The solid oxide fuel cell (SOFC) is considered for use for electrical power generation in aviation applications largely due to its high potential fuel-to-electricity conversion efficiency [26,27]. But there are numerous issues in such aircraft applications, some of which follow:

- The storage of pure hydrogen (as fuel for the SOFC) in the gas phase requires very large storage tanks, which is a significant disadvantage for mobile applications. So, the hydrogen should be stored as either a compressed gas or a liquid. Storage of hydrogen as a gas typically requires very high pressures (350–700 bar), while the storage of hydrogen as a liquid requires cryogenic temperatures (−252.8 °C at atmospheric pressure). However, storage of hydrogen in either liquid or gaseous form can lead to technical difficulties and additional costs.

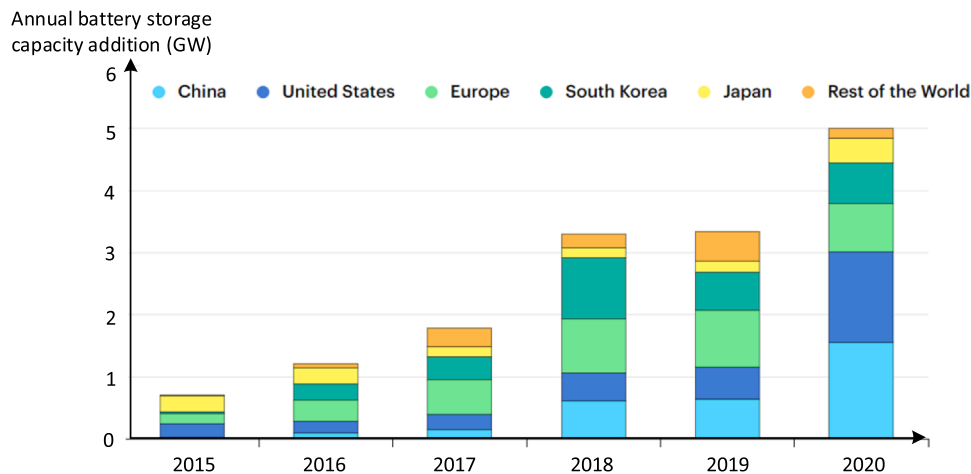


Fig. 1. Annual energy storage additions by country from 2015 to 2020. Modified from [1].

- The challenges described in the last point regarding the direct use of pure hydrogen in electric aircraft with a SOFC have led to efforts to find alternative hydrogen storage mediums like fuels containing hydrogen. For instance, kerosene, ammonia, methane, ethanol and gasoline are other fuels that can be used in a SOFC. The use of such fuels requires reformation at high temperatures. A reformer is a device that produces hydrogen from a source that contains hydrogen. For short-chain hydrocarbon fuels like methane, natural gas and ethanol, fuel reformation can take place internally, avoiding the need for an external reformer.
- In addition, a supplemental battery system is often used in an electric vehicle to provide electrical power during takeoff and climb when the need for power is relatively high. One of the main challenges in the application of batteries in aviation applications is the effect of temperature on battery performance. Therefore, a cooling process for the battery pack in electric aircraft is necessary to maintain a uniform temperature within the desired range for effective and long-term operation of battery cells.

Increasing research is being undertaken by NASA on developing and evaluating, for aircraft applications, means for the use of alternative fuels like methane in advanced SOFCs containing internal reformers, and fuel/energy storage technologies [27]. Studies are being carried out to identify concepts with high efficiency and payoff potential, and related technologies, for further development of electric aircraft. The study considered here can support that activity.

Relatively few studies on integrated battery and fuel cell systems in electric aircraft have been reported. As well, there are no studies reported on the integration of a battery thermal management system with other units in which the generated heat in the battery is transferred to the cooling medium and recovered in another unit like a SOFC with an internal reformer. In this study, we consider a novel hybrid SOFC-battery-gas turbine propulsion system which supplies electrical power for an electric aircraft. This hybrid system incorporates a lithium ion battery pack, a SOFC with an internal reformer, compressors, a humidifier, heat exchangers, a gas turbine/generator and other components. Proposing the integration of a battery thermal management system with a SOFC is motivated by the fact that preheating the input water before it is pumped to the internal reformer can potentially improve its performance. The heat generated in the battery is removed through convection to the liquid water flow that maintains the battery temperature in the desired operating range. Also, simulating the SOFC and lithium ion battery systems is important to investigate the compatibility and applicability of the proposed integrated system for electric aircraft. System component models are developed to evaluate the performance for the system design, and the components and system are analyzed. Heat generation rates in lithium ion battery cells are determined through thermal analyses of batteries, where the battery temperature is allowed to vary with time. Average values for heat generation rates in lithium ion battery cells are used in simulations of the remaining system components. The electrical power produced by the SOFC with an internal reformer is determined with an electrochemical model and, finally, the performance of the overall hybrid system is assessed in terms of its energy efficiency. The effects of key operating parameters on battery voltage, temperature and heat generation rate (in the battery), as well as voltage, power density and energy efficiency of the integrated SOFC-battery-gas turbine system, are analyzed and discussed.

2. System description

In the study, the design of the hybrid-electric aircraft utilizes a propulsion system in which a SOFC fueled by methane and a gas turbine provide the majority of the electrical power. The lithium ion batteries are used to supply electrical power during high power load operating conditions.

Fig. 2 shows the SOFC-battery-gas turbine system. It includes the following components: air and methane compressors, water pumps, battery pack, heat exchangers, a mixer, a SOFC with an internal reformer, an afterburner, a gas turbine and a motor. The SOFC reactants including water, methane and air are pressurized and preheated in the heat exchangers where their temperature increases to 1000 K.

High battery temperatures shorten its lifetime by increasing the rate of the aging processes within the battery. This makes an effective and efficient cooling necessary for high power applications. In the system considered in the study, water is pumped to the lithium ion battery pack to remove part of the heat generated in the battery and to cool the individual cells by passing around them. Excess water exiting the battery pack is discarded from the system while the main portion of water is pumped to the heat exchanger where it is converted to steam and then mixed with high temperature methane in the mixer before entering the SOFC. In the SOFC, which is equipped with an internal reformer, the mixture of steam and methane is first converted to the carbon monoxide and hydrogen through the steam reforming reaction, and then converted to carbon dioxide and hydrogen through the water-gas shift reaction. The produced gases are transferred to the anode side of the SOFC where the hydrogen gas reacts with the oxygen of in the high temperature air at the cathode to produce water.

The unconsumed fuel, excess air and products of the SOFC-internal reformer are utilized in the afterburner for generation of high temperature streams. These are used in the gas turbine for power generation as well as in the heat exchangers for heating the SOFC reactants. The combustion gases exiting the gas turbine can be further utilized for preheating the inlet air to the aircraft. The electrical power produced by the SOFC and gas turbine is used by the aircraft motor, and this usage is assumed constant throughout the flight. Additional power produced during the flight by the gas turbine and SOFC can be stored in the lithium ion battery and utilized later during takeoff or climbing when much greater electrical power is required.

3. Modeling and analysis

3.1. Electrochemical model of lithium ion battery

Each battery cell in the pack includes negative and positive porous electrodes which are separated by a polymer separator. The porous electrodes consist of an insertion electrode material (solid phase), a polymer electrolyte (solution phase) and a conductive filler that provides additional electronic conductivity. A carbon material and lithium manganese oxide (LiMn_2O_4) are used as active materials for the negative and positive electrodes, respectively. A pseudo-homogeneous mixture is used for modeling the porous electrodes. Details on the thermophysical behavior and properties of the porous electrodes and the polymer electrolyte during discharge and charge of the battery cell are provided elsewhere [28,29].

The battery pack is comprised of cylindrical lithium ion batteries which are cooled through the passage of water between the batteries inside the pack. Fig. 3 (a) shows half of a vertical cross section of a cylindrical lithium ion battery. It contains a mandrel (isolator around which the battery cell sheets are wound), a canister and an active material domain (wound sheets of cell material) including negative and positive electrodes and a separator. The height of the battery cell is 65 mm and the radius is 9 mm. The thickness of the steel canister is 0.25 mm and the radius of the mandrel is 2 mm. For a cylindrical battery, as shown in Fig. 3, the separator, anode and cathode sheets are cut into two long strips which are wound on a cylindrical mandrel to form a cylindrical structure. In the present paper, a two-dimensional and axially symmetric model is developed for the lithium ion battery. The reason for considering an axial-symmetric condition is that heat conduction in the spiral direction (direction through which the anode, cathode and separator layers are rolled around the mandrel) can be neglected for a spirally wound battery cell (see Fig. 3 (b)).

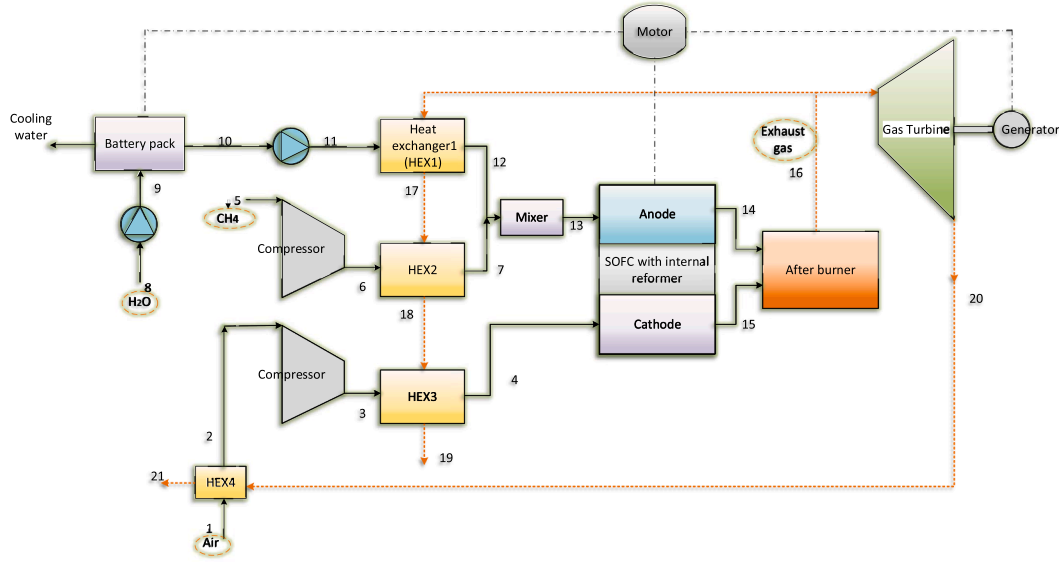


Fig. 2. Flow diagram for a hybrid-electric aircraft energy system consisting of a methane-fueled SOFC, a gas turbine and lithium ion batteries for supplying electrical power.

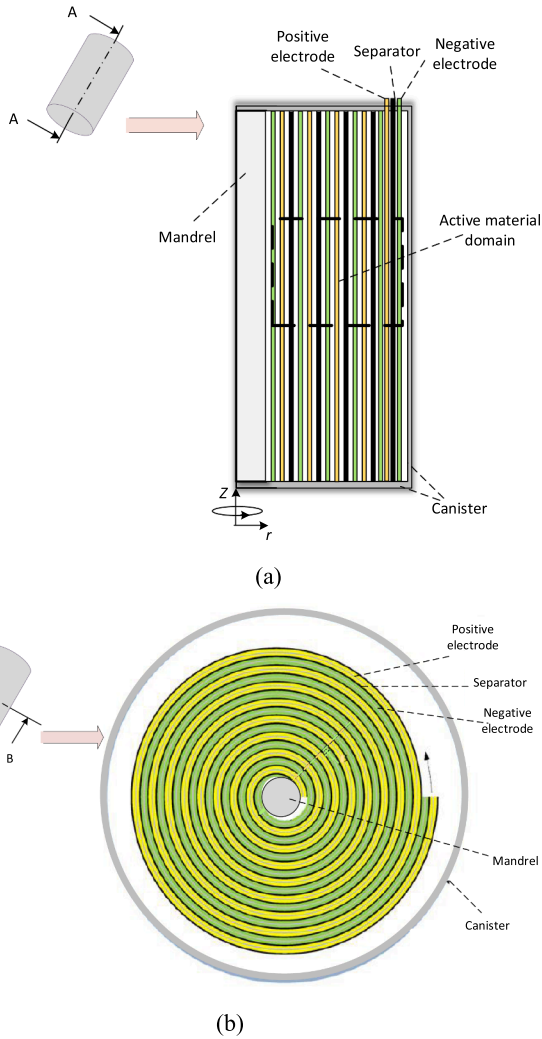


Fig. 3. (a) Vertical cross section of half of an upright cylindrical lithium ion battery. The center of the battery is at the left. (b) Cross section of a cylindrical lithium ion battery. A spiral roll is composed of anode, cathode and separator.

The variation with time of the salt concentration in the electrode solution phase (i.e., polymer electrolyte) of the porous electrode can be expressed based on a material balance for the lithium in the polymer/salt phase, as follows [28], [30] and [24]:

$$\epsilon \frac{\partial c_s}{\partial t} = (\nabla \cdot (D_s \nabla c_s) + \bar{a} J_n (1 + t_+^0)) - \frac{i_2 \nabla t_+^0}{F} \quad (1)$$

where c_s , ϵ , t , D_s , F and ∇t_+^0 respectively denote salt concentration, volume fraction of polymer, time, salt diffusion coefficient, Faraday constant and transference number. The transference number indicates that the flow of ions creates an ionic current in an electrolyte. This means that each ion in the electrolyte contributes to the ionic current by carrying a fraction in the solution. The terms i_2 , $\bar{a}$ and J_n respectively denote ionic current density in the solution phase, specific interfacial area and the pore wall flux of lithium ions. These terms are related as follows:

$$\bar{a} = \frac{1}{F} \nabla \frac{i_2}{J_n} \quad (2)$$

where the ionic current density in the electrolyte phase of the electrode can be expressed as:

$$i_2 = -\ddot{A}_s^{eff} \nabla \Phi_2 + \left(\frac{2\ddot{A}_s RT}{F} \left(1 + \frac{\partial \ln \gamma_{\pm}}{\partial \ln c_s} \right) (1 - t_+^0) \right) \nabla \ln c_s \quad (3)$$

where $\ddot{A}_s^{eff}$ denotes the effective electrolyte ionic conductivity, R the universal gas constant, T the absolute temperature of the cell, $\gamma_{\pm}$ the activity coefficient and Φ_2 the electrical potential in the solution phase. Note in general that subscript 1 indicates the solid phase and subscript 2 the electrolyte phase of the composite electrode. The parameter $\ddot{A}_s^{eff}$ can be determined based on the ionic conductivity of electrolyte ($\ddot{A}_{s,0}$) and the phase volume fraction (ϵ) as

$$\ddot{A}_s^{eff} = \ddot{A}_{s,0} \epsilon^\gamma \quad (4)$$

where γ is the Bruggeman coefficient, which is taken here to be 3.3 [28]. The ionic conductivity of the electrolyte depends on the cell temperature, salt concentration and solvent ratio of ethylene carbonate (EC) to dimethyl carbonate (DMC). The electrolyte used in this study is comprised of 2 M lithium hexafluorophosphate (LiPF_6) salt in a 1:2 EC: DMC (by volume) solvent and a porous membrane comprised of poly

vinylidene fluoride-hexafluoropropylene (PVDF-HFP).

The insertion electrode is made up of spherical particles with a radius of R_s . The lithium concentration in the electrode solid matrix can be defined based on the diffusion of lithium ions into the solid particles as follows:

$$\frac{\partial c_{Li}}{\partial t} = D_{Li} \left(\frac{\partial^2 c_{Li}}{\partial r^2} + \frac{2}{r} \left(\frac{\partial c_{Li}}{\partial r} \right) \right) \quad (5)$$

where D_{Li} is the diffusion coefficient of lithium in the solid phase of the composite electrode and r is the dimension through the radius (radial direction) of the particles. One of the boundary conditions for Eq. (5) is no flux of lithium ions across at the center of solid particles (i.e., $\partial c_{Li} / \partial r = 0$ at $r = 0$). The other boundary condition, which relates the solid and polymer electrolyte phases of the porous electrode on the outer surface of electrode, is based on the relation between the lithium ion flux into the particles and the rate of lithium ion diffusion to the surface of the particles, i.e.,

$$\frac{\partial c_{Li}}{\partial t} \Big|_{r=R_s} = \frac{J_n}{D_{Li}} \quad (6)$$

In the polymer separator, Eq. (1) is applied with $J_n = 0$ (no pore wall flux of lithium ions) and $\epsilon = 1$ (the entire separator is a polymer so the polymer fraction is 1). Thus, the variation with time of the salt concentration in the polymer separator can be expressed as:

$$\frac{\partial c_s}{\partial t} = (\nabla \cdot (D_s \nabla c_s)) - \frac{i_2 \nabla t_+^0}{F} \quad (7)$$

Ionic current density is conserved in the two regions of the separator and the electrolyte phase of porous electrode (i.e., i_2 (in separator) = i_2 (in electrolyte phase of electrode)). Based on Eq. (3), the electrical potential variation in the separator can be written as:

$$\nabla \Phi_2 = \frac{i_2}{\bar{A}_{s,eff}} + \frac{1}{\bar{A}_{s,eff}} \left(\frac{2\bar{A}_{s,eff} RT}{F} (1 + \frac{2\bar{A}_{s,eff} RT}{F}) \frac{\partial \ln \gamma_{\pm}}{\partial \ln c_s} \right) (1 - t_+^0) \nabla \ln c_s \quad (8)$$

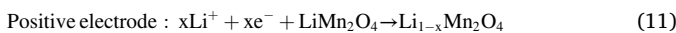
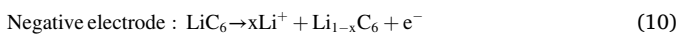
The first term on the right side represents the ohmic overpotential due to the ionic resistivity of the electrolyte and the second term represents the concentration overpotential due to a concentration gradient within the polymer separator and solution phase of the porous electrode.

The boundary condition for the electrical potential at the polymer separator is no potential change in the solid phase, as follows:

$$\frac{\partial \Phi_1}{\partial x} \Big|_{x=\delta_- + \delta_s} = 0 \quad (9)$$

Other boundary conditions considered in the model of the lithium ion battery cell are (i) a potential of zero for the current collector at the negative electrode and (ii) at the positive electrode current collector, the current density is specified as a finite constant through discharge and charge and zero through the open circuit duration.

The surface overpotential is the resistance to electrochemical reactions in the negative and positive porous electrodes. In fact, a driving force (activation energy) is required to induce the electrochemical reactions to occur. During discharge, the reactions at the negative and positive electrodes can be expressed as follows:



It can be assumed that the reactions at the electrodes follow Butler-Volmer kinetics. The general expression of the Butler-Volmer equation can be written as:

$$i = i_{0,1} \left(\exp \left(\frac{\alpha_- \eta_{s1} n_e F}{RT} \right) - \exp \left(\frac{-\alpha_+ \eta_{s1} n_e F}{RT} \right) \right) \quad (12)$$

where i is the net current density. Note that, due to different rates for the positive and negative electrodes reactions, a net current density is

produced in the cell. The terms α_+ and α_- respectively are the transfer coefficients of reaction at the positive and negative electrodes. These are 0.5 for both reactions at the positive and negative electrodes. Also, $i_{0,1}$ is the exchange current density in the solid phase of electrode, which can be expressed as follows:

$$i_{0,1} = F k_{+1}^{a_{+1}} k_{-1}^{a_{-1}} (c_{max} - c)^{a_{+1}} c^{a_{-1}} \quad (13)$$

Here, c_{max} and c are the solubility limit of the lithium salt and electrolyte concentration, respectively, and k_+ and k_- are rate constants for the reactions occurring at the positive and negative electrodes, respectively.

The term η_{s1} in Eq. (12) is the local surface overpotential, which is equivalent to the difference between the positive/negative electrode potentials ($\Phi_1 - \Phi_2$) and the open circuit potential ($E_{OCP,1}$). That is,

$$\eta_{s1} = E_{OCP,1} - (\Phi_1 - \Phi_2) \quad (14)$$

where Φ_1 and Φ_2 respectively are the electrical potentials of the solid and solution phases of the composite electrode, respectively. If a lithium foil is used for the negative electrode and it is considered the reference electrode, the open circuit potential for the lithium foil electrode is zero. For this study, in which porous composite electrodes are used for both the negative and positive electrodes, the Butler-Volmer equation becomes:

$$j_n = k(c_{1,max} - c_{1,Li})^{0.5} c_{1,Li}^{0.5} \left[\exp \left(-\frac{F}{2RT} (\eta_s) \right) - \exp \left(\frac{F}{2RT} (\eta_s) \right) \right] \quad (15)$$

where $c_{1,max}$ and $c_{1,Li}$ are the maximum concentration of lithium in the solid phase of the composite electrode and the concentration of lithium in the solid phase of the composite electrode. The term k is the reaction rate constant for the charge transfer reaction at the electrode surface (i.e., the electrode-polymer separator interface). The reaction rate constant is k_+ for the positive electrode surface and k_- for the negative electrode surface. The electrode surface overpotential η_s can be written as

$$\eta_s = E_{OCP} - (\Phi_1 - \Phi_2) = E_{eq} \quad (16)$$

where $\Phi_{+/-}$ is the positive/negative electrode potential, which is the difference between the corresponding electrical potentials of all phases existing at the positive/negative electrode (i.e., $\Phi_{+/-} = (\Phi_1 - \Phi_2)$).

Doyle et al. [28] proposed empirical correlations for the open circuit potential of the negative electrode composed of carbon material and for the positive electrode composed of lithium manganese oxide (LiMn_2O_4) for a very low discharge rate and a battery temperature of 20 °C. That is,

$$E_{eq-} = -0.16 + 1.32 \exp(-3x) + 10 \exp(-2000x) \quad (17)$$

$$E_{eq+} = 4.2 + 0.06 \tanh[-14.56y + 8.61] - 0.03 \left[\frac{1}{(1-y)^{0.5}} - 1.9 \right] - 0.16 \exp(-0.05y^8) + 0.81 \exp(-40(y - 0.14)) \quad (18)$$

where x and y are stoichiometric coefficients for the carbon electrode (LiC_6) and lithium manganese oxide (LiMn_2O_4).

The superficial current density in the electrode solid phase can be expressed by Ohm's law as follows:

$$i_1 = -\sigma \nabla \Phi_1 \quad (19)$$

where σ is the electronic conductivity, which indicates the resistance to electron transfer in the solid phase of the negative/positive electrodes, and $\nabla \Phi_1$ denotes the ohmic overpotential due to electron transfer.

From the conservation of current in two phases (i.e., $\nabla(i_1 + i_2) = 0$), the superficial current density is equal to the summation of the current densities at the solid and solution phases. That is,

$$i = i_1 + i_2 \quad (20)$$

Finally, the cell potential and current respectively can be written as

$$I_B = Ai \quad (21)$$

$$\begin{aligned} V_B &= \Phi_+ - \Phi_- \\ &= E_{OCF} + (\eta_{s+} + \eta_{s-}) + (\nabla\Phi_{2,+} + \nabla\Phi_{2,-} + \nabla\Phi_{2,sep}) + (\nabla\Phi_{1,+} + \nabla\Phi_{1,-}) \end{aligned} \quad (22)$$

where A is the battery surface area, and Φ_+ and Φ_- respectively are the electrical potentials of the positive and negative electrodes. The term $(\eta_{s+} + \eta_{s-})$ denotes the surface overpotential at the positive and negative electrodes, $(\nabla\Phi_{2,+} + \nabla\Phi_{2,-} + \nabla\Phi_{2,sep})$ is the concentration overpotential and ohmic overpotential due to ionic transfer in the electrolyte at the positive and negative electrodes and the separator, and $(\nabla\Phi_{1,+} + \nabla\Phi_{1,-})$ is the ohmic overpotential due to electron transfer in the positive and negative electrodes. The parameter details and values used in the development of the electrochemical model of lithium ion battery are listed in Table 1.

3.2. Thermal model of lithium ion battery

In this study, the battery temperature is allowed to vary with time (i.e., we have a non-isothermal condition). Also, since the thermal conductivities of the components of the lithium ion battery cell are relatively high and the heat generated in the cell is not excessive, it can be assumed that the battery has a uniform temperature profile. In addition, when the chemistry of the battery is not highly affected by small changes in the battery temperature, using a lumped parameter approach which is based on the average temperature can predict the battery thermal behavior with adequate accuracy. In the rest of this section, the thermal model of a lithium ion battery cell is presented and applied to determine the battery temperature variations over time. The developed electrochemical and thermal models are coupled through a multiphysics coupling node in the Comsol software to connect the heat generated in the battery and the temperature of the battery.

The general energy equation for a multi-component system (like a battery), which is derived from a differential energy balance, can be written as:

$$\frac{\partial(\rho_{p,B}c_{p,B}T_B)}{\partial t} = \nabla \cdot k \nabla T + \dot{Q}_{gen} \quad (23)$$

where $c_{p,B}$, ρ_B , T_B and t respectively denote specific heat at constant

Table 1

Parameter details and values used in the development of the electrochemical model of the lithium ion battery.

Solid phase lithium diffusivity for negative electrode	$3.9 \times 10^{-14} \text{ m}^2/\text{s}$
Solid phase lithium diffusivity for positive electrode	$1 \times 10^{-13} \text{ m}^2/\text{s}$
Particle radius negative electrode	$1.25 \times 10^{-5} \text{ m}$
Particle radius positive electrode	$8 \times 10^{-6} \text{ m}$
Electrolyte phase volume fraction for positive electrode	0.63
Electrode phase volume fraction for positive electrode	0.297
Initial electrolyte salt concentration	2050 mol/m ³
Electrolyte phase volume fraction for negative electrode	0.503
Solid phase volume fraction for negative electrode	0.471
Transport number	0.363
Diffusion coefficient for free stream electrolyte	$7.5 \times 10^{-11} \text{ m}^2/\text{s}$
Maximum solid phase concentration for negative electrode	26390 mol/m ³
Initial concentration for negative active electrode material	14880 mol/m ³
Initial concentration for positive active electrode material	3920 mol/m ³
Electrolyte salt concentration	1000 mol/m ³
Solid phase conductivity for negative electrode	100 S/m
Reference exchange current density for negative electrode	11.71 A/m ²
Reference exchange current density for positive electrode	11.14 A/m ²
Reaction rate coefficient for positive electrode	0.5
Reaction rate coefficient for negative electrode	0.5
Bruggeman coefficient	3.3

Data from [28], [34], [24] and [29].

pressure, density of battery, battery temperature and time. Also, $\dot{Q}_{gen}$ is the heat generation rate per unit volume for the battery.

For a two-dimensional model of a cylindrical battery cell considered here, the above equation can be written as

$$\frac{\partial(\rho_B c_{p,B} T_B)}{\partial t} = \frac{1}{r} \frac{\partial}{\partial r} \left(k_r r \frac{\partial T_B}{\partial r} \right) + \frac{\partial}{\partial z} \left(k_z \frac{\partial T_B}{\partial z} \right) + \dot{Q}_{gen} \quad (24)$$

where k_r and k_z respectively are the thermal conductivities of the cell in the radial and axial directions. For a single cell, which has a small thickness, the *lumped-parameter approach* is applicable. This indicates the condition where the ratio of convective heat transfer coefficient (h_{conv}) times the thickness of the cell (δ_B) to the thermal conductivity of the cell (k) is very small (i.e., $h_{conv}\delta_B/k \ll 1$). Hence, the assumption of uniform temperature throughout the cell is reasonably accurate.

The heat generation rate in the battery is due to heat of electrochemical reactions (reversible heat) and internal resistances (irreversible heat) within the cell. A general expression for the battery heat generation rate per volume of the cell can be written as:

$$\dot{Q}_{gen} = \frac{1}{V_B} (I_B (E_{OCF} - E_B) - I \left(T_B \frac{\partial E_{eq}}{\partial T} \right)) \quad (25)$$

where E_{eq} , I_B , E_B and V_B respectively denote equilibrium cell potential, cell current, cell voltage and volume of cell. The reversible heat rate term in Eq. (25) (i.e., $I(T_B \partial E_{eq} / \partial T)$) is equivalent to $T_B \Delta S I_B / n_e F$ in which ΔS is the entropy change due to electrochemical reactions occurring in the cell and n_e is the number of electrons in the balanced electrochemical half reaction.

Each battery is comprised of three layers (anode, cathode and separator) which have different thermal conductivities in different directions of the cylindrical battery cell. The battery overall thermal conductivities in radial (k_r) and axial (k_z) directions respectively can be expressed as:

$$k_r = \frac{\sum \delta_i}{\sum (\delta_i / k_i)} \quad (26)$$

$$k_z = \frac{\sum \delta_i k_i}{\sum (\delta_i)} \quad (27)$$

where δ_i and k_i are the thickness of layer i and thermal conductivity of layer i . Different values for these parameters apply to the different materials constituting the layers, respectively.

Similarly, the density and specific heat at constant pressure of the battery with its multi-layer structure can be written respectively as:

$$\rho_B = \frac{\sum \delta_i \rho_i}{\sum (\delta_i)} \quad (28)$$

$$c_{p,B} = \frac{\sum \delta_i c_{p,i}}{\sum (\delta_i)} \quad (29)$$

where ρ_i and $c_{p,i}$ are the density and specific heat capacity of layer i in the battery cell. The water that is used for the SOFC first is passed through the lithium ion battery pack for cooling purposes and then is pumped through the heat exchanger. To account for the cooling effect of water between the battery cells, a heat flux boundary condition is specified on the battery canister surface. Doing so allows the convective heat transfer between the cooling water and the battery surface to be evaluated. The input parameters for the transient model for the lithium ion battery are listed in Table 2, along with their values.

3.3. Solid oxide fuel cell (SOFC) model

The SOFC is a high-temperature (600 to 1000 °C), all-ceramic and solid-state unit in which an oxide ion conducting ceramic material is used as the electrolyte. A SOFC has potential advantages over proton

Table 2
Input design parameters for transient model of the lithium ion battery cell.

Parameter	Value
Thickness of battery canister	2.5×10^{-4} m
Battery radius	9×10^{-3} m
Battery height	6.5×10^{-2} m
Mandrel radius	2×10^{-3} m
Length of negative electrode	5.5×10^{-5} m
Length of separator	3×10^{-5} m
Length of positive electrode	5.5×10^{-5} m
Thickness of negative current collector	7×10^{-6} m
Thickness of positive current collector	1×10^{-5} m
Cell thickness	1.57×10^{-4} m
Thermal conductivity of positive electrode	1.58 W/(m·K)
Thermal conductivity of negative electrode	1.04 W/(m·K)
Thermal conductivity of positive current collector	170 W/(m·K)
Thermal conductivity of negative current collector	398 W/(m·K)
Thermal conductivity of separator	0.344 W/(m·K)
Density of positive electrode	2328.5 kg/m ³
Density of negative electrode	1347.3 kg/m ³
Density of positive current collector	2770 kg/m ³
Density of negative current collector	8933 kg/m ³
Density of separator	1009 kg/m ³
Heat capacity of positive electrode	1269.2 J/(kg·K)
Heat capacity of negative electrode	1437.4 J/(kg·K)
Heat capacity of positive current collector	875 J/(kg·K)
Heat capacity of negative current collector	385 J/(kg·K)
Heat capacity of separator	1978.2 J/(kg·K)
Battery angular thermal conductivity	29.557 W/(m·K)
Battery radial thermal conductivity	0.89724 W/(m·K)
Battery density	2055.2 kg/m ³
Battery heat capacity	1399.1 J/(kg·K)
Inlet temperature	298.15 K
Initial temperature	298.15 K
Cell capacity	14,400C
Ohmic overpotential at 1C	0.0045 V
C-rate	2

Data from [35,36,37].

exchange membrane (PEM) fuel cells for aircraft applications. For instance, the relatively higher operating temperature of a SOFC results in greater opportunities to extract waste heat from the SOFC product streams that can be used for other processes in the system such as fuel heating and reformation.

The model and analysis of the SOFC is carried out here under equilibrium and steady state conditions. The following assumptions are invoked, following [31], for the methane-fueled SOFC equipped with an internal reformer:

- Air is composed of 21 % oxygen and 79 % nitrogen on a volume basis.
- The temperature difference between the SOFC reactants and products is fixed at 50 K.
- Radiation heat transfer between the solid structures and gas channels is negligible.
- Unused fuel that is not consumed in the SOFC is taken to be oxidized completely in the afterburner.
- The solid oxide fuel cell is insulated fully, so there is no heat interaction with the environment.
- Contact resistances can be neglected.
- Only H₂ participates in the electrochemical reaction with oxygen, and carbon monoxide is converted to carbon dioxide and hydrogen through the water–gas shift reaction.
- Chemical equilibrium is taken to exist for the gas mixture at the exit of the fuel channel.

For a methane-fueled solid oxide fuel cell, it is more economical to use an internal reformer rather than an external reformer. Fig. 4 shows an internal reformer combined with a solid oxide fuel cell. In the reforming of methane with steam, the main chemical reactions are as follows:



The steam reforming reaction in Eq. (30) is slow and endothermic while the water–gas shift reaction in Eq. (31) is fast and exothermic. The heat interaction required for steam reforming is much higher than for the water–gas shift reaction (i.e., at a reaction temperature of 1300 K, the steam reforming enthalpy of reaction is 227.5 kJ/mol and for the water–gas shift reaction is −31.8 kJ/mol). So, the overall reaction in the internal reformer is endothermic. The hydrogen produced in the steam reforming and water–gas shift reactions is used for the electrochemical reaction in the SOFC. The overall reaction at the SOFC (i.e., H₂ + O₂ → H₂O) is exothermic. So, the generated heat by the SOFC can be used to supply heat for steam reforming in the reformer. This reduces the dependency of the internal reformer-SOFC system on a cooling system. The temperature difference between the reactants at the SOFC inlet and the products at the SOFC outlet is assumed here to be 50 K. In addition, in the internal reformer-SOFC system, the unused fuel can be utilized in the afterburner to produce heat for the heat exchangers that heat the SOFC reactants, as well as in the gas turbine for electrical power production.

In the remainder of this subsection, a zero-dimensional model is developed for SOFC which is assumed to operate at steady state condi-

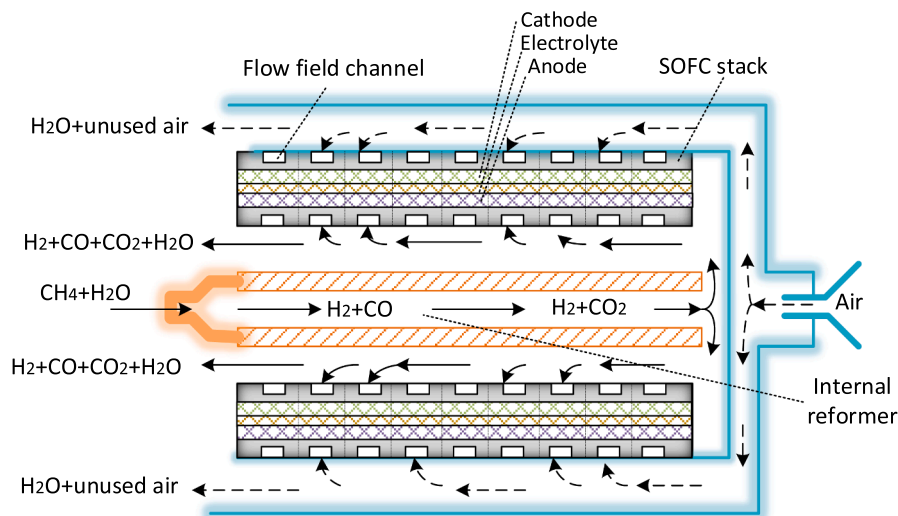


Fig. 4. Solid oxide fuel cell with internal reformer.

tions. Like the PEM fuel cell that is modeled in case study 1, the output voltage of the SOFC can be expressed as:

$$E_{SOFC} = E_{OCP,SOFC} - ((\eta_{a,a} + \eta_{a,c}) + (\eta_{c,a} + \eta_{c,c}) + \eta_{ohm}) \quad (32)$$

where $E_{OCP,SOFC}$ is the open circuit potential of the SOFC, $\eta_{a,a}$ and $\eta_{a,c}$ are the activation overpotentials at the anode and cathode respectively, $\eta_{c,a}$ and $\eta_{c,c}$ are the concentration overpotentials at the anode and cathode respectively and η_{ohm} is the ohmic overpotential. The calculations of the equilibrium potential and the voltage losses in the SOFC are presented and described in Table 3.

At the equilibrium condition, the molar conversion rates for the reforming and shift chemical reactions in the internal reformer are denoted r_1 and r_2 , while the molar conversion rate for the overall electrochemical reaction in the SOFC is denoted r_3 . These rates can be evaluated using the equilibrium constant and current relations. The equilibrium constant for the shift reaction can be written as

$$\ln K_s = -\frac{\Delta \bar{g}_s}{RT_{SOFC}} = \ln \left[\frac{r_2(3r_1 + r_2 - r_3)}{(r_1 - r_2)(1.5r_1 - r_2 + r_3)} \right] \quad (33)$$

where K_s is the equilibrium constant for the reaction. The specific Gibbs free energy can be given as:

$$\Delta \bar{g}_s = \bar{g}_{s,H_2} + \bar{g}_{s,CO_2} - \bar{g}_{s,H_2O} - \bar{g}_{s,CO} \quad (34)$$

Here the specific Gibbs free energy condition can be determined as $\bar{g}_s = \bar{h} - T_{SOFC}\bar{s}$ where T_{SOFC} is the temperature of the SOFC products. The molar enthalpy ($\bar{h}$) is evaluated at T_{SOFC} and the molar entropy ($\bar{s}$) at T_{SOFC} and standard pressure (101 kPa). Based on the Faraday law, the molar consumption rate of methane at the anode can be written as

$$\dot{N}_{CH_4,consumed} = \frac{i_{SOFC} N_{SOFC} A_{SOFC}}{2F} \quad (35)$$

where i_{SOFC} is the net exchange current density of the solid oxide fuel cell, N_{SOFC} the number of fuel cells in the stack, A_{SOFC} the active area of the cell and F the Faraday constant.

The fuel utilization factor can be expressed as:

$$U_F = \frac{\dot{N}_{CH_4,consumed}}{\dot{N}_{CH_4,in}} \quad (36)$$

where $\dot{N}_{CH_4}$ is molar flow rate of methane. The molar flow rate of working fluids in the SOFC system with the internal reformer can be calculated through a mass balance. Table 4 lists the mass balances for the streams of the examined system. In addition, the input design

Table 4

Molar flow rate equations for the SOFC with internal reformer.

Description	Equations
Chemical reactions in the SOFC with internal reformer and the corresponding molar conversion ratio	$r_1[CH_4 + H_2O \rightarrow CO + 3H_2]r_2[CO + H_2O \rightarrow CO_2 + H_2]r_3[H_2 + 0.5O_2 \rightarrow H_2O]$
Mass balance in the SOFC with internal reformer	$\dot{n}_{CH_4,13} = r_1\dot{n}_{H_2,0,13} = x_{sc}\dot{n}_{CH_4,13} =$ $2.5r_1x_{sc} = \dot{n}_{H_2,0,13}/\dot{n}_{CH_4,13}\dot{n}_{H_2,14} = 3r_1 +$ $r_2 - r_3\dot{n}_{CO,14} = r_1 - r_2\dot{n}_{CO,14} = r_2\dot{n}_{H_2,0,14} =$ $\dot{n}_{H_2,0,13} - r_1 - r_2 + r_3 = 1.5r_1 - r_2 +$ $r_3\dot{n}_{O_2,4} = r_3/2U_{air}$ utilization ratio (U_a) = $\frac{consumed O_2}{supplied O_2} = \frac{(r_3/2)}{\dot{n}_{O_2,4}}$ $\dot{n}_{N_2,15} = \dot{n}_{N_2,4}\dot{n}_4 = \dot{n}_{O_2,4} + \dot{n}_{N_2,4}\dot{n}_{15} =$ $\dot{n}_{O_2,15} + \dot{n}_{N_2,15}\dot{n}_{14} = \dot{n}_{H_2,14} + \dot{n}_{H_2,0,14} +$ $\dot{n}_{CO,14} + \dot{n}_{CO_2,14}\dot{n}_{13} = \dot{n}_{H_2,0,13} + \dot{n}_{CH_4,13}$

parameters used to simulate the SOFC with the internal reformer are presented in Table 5.

3.4. Energy analysis of the system

A general energy rate balance for components of the system under

Table 5

Input design parameters considered for the SOFC/internal reformer model.

Parameter	Value
U_F	85 %
A_{SOFC}	100 cm ²
i_{SOFC}	0.8 A/cm ²
i_{oa}	0.56 A/cm ²
i_{oc}	0.25 A/cm ²
$D_{a,eff}$	0.2×10^{-4} m ² /s
$D_{c,eff}$	0.5×10^{-5} m ² /s
δ_a	0.5×10^{-3} m
δ_c	0.5×10^{-4} m
δ_e	0.1×10^{-4} m
δ_{int}	0.3×10^{-2} m
x_{sc} (steam to carbon ratio)	2.5
SOFC pressure drop	2 %
Number of cells in a stack	100
Number of stacks	34

Data from [40,39].

Table 3

Equations for determination of equilibrium potential and overpotentials in the SOFC.

Potential type	Equation	Description
Equilibrium potential, $E_{OCP,SOFC}$	$E_{OCP,SOFC} = -\frac{\Delta \bar{g}}{2F} + \frac{RT_{SOFC}}{2F} \ln \left(\frac{a_{H_2,14} a_{O_2,15}^{0.5}}{a_{H_2,0,14}} \right) a_{H_2,0,14} = \frac{P_{H_2,0,14}}{P_{ref}} = \frac{x_{H_2,0,14} P_{14}}{P_{ref}} a_{H_2,14} =$ $\frac{P_{H_2,14}}{P_{ref}} = \frac{x_{H_2,14} P_{14}}{P_{ref}} a_{O_2,15} = \frac{P_{O_2,15}}{P_{ref}} = \frac{x_{O_2,15} P_{15}}{P_{ref}} \Delta \bar{g} = \bar{g}_{H_2,0} - \bar{g}_{H_2} - 0.5 \bar{g}_{O_2} \bar{g} =$ $\bar{h} - T_{SOFC} \bar{s}$	Parameters a_i and x_i respectively are the activity factor of species i in the electrochemical reaction and the molar concentration of species i . Also, $\bar{g}$, $\bar{h}$ and $\bar{s}$ denote molar Gibbs free energy, molar enthalpy and molar entropy, respectively. P_i denotes partial pressure of species i .
Activation overpotential, η_a	$\eta_a = \eta_{a,a} + \eta_{a,c} \eta_{a,a} = \frac{RT_{SOFC}}{F} (\sinh^{-1} \left(\frac{i}{2i_{oa}} \right)) \eta_{a,c} = \frac{RT_{SOFC}}{F} (\sinh^{-1} \left(\frac{i}{2i_{oc}} \right))$	Parameters i , i_{oa} and i_{oc} denote current density, exchange current density of anode and exchange current density of cathode, respectively. T_{SOFC} and R respectively are SOFC temperature and universal gas constant.
Ohmic overpotential, η_{ohm}	$\eta_{ohm} = (R_c + \sum j \delta_j) i \rho_{int} = 1/(9.3 \times 10^6 \exp(-1100/T_{SOFC})) \rho_c = 1/(42 \times 10^4 \exp(-1200/T_{SOFC})) \rho_e = 1/(3.34 \times 10^4 \exp(-10300/T_{SOFC})) \rho_a = 1/(95 \times 10^6 \exp(-1150/T_{SOFC}))$	Parameters ρ_{int} , ρ_c , ρ_e and ρ_a denote electrical resistivity of interconnect, cathode, electrolyte and anode, respectively. Also, δ_i is the thickness of layer i of the SOFC.
Concentration overpotential, η_c	$\eta_c = \eta_{c,a} + \eta_{c,c} \eta_{c,a} = \frac{RT_{SOFC}}{2F} \left(\ln \left(1 + \frac{P_{H_2} i}{2P_{H_2,0} i_{as}} \right) - \ln \left(1 - \frac{i}{i_{la}} \right) \right) \eta_{c,c} =$ $-\frac{RT_{SOFC}}{4F} \ln \left(1 - \frac{i}{i_{lc}} \right) j_{as} = 2FP_{H_2} D_{a,eff} / (RT_{SOFC} \delta_a) j_{ac} = 4FP_{O_2} D_{c,eff} /$ $\left(\frac{P_{15} - P_{O_2,15}}{P_{15}} (RT_{SOFC} \delta_c) \right)$	Parameters $\eta_{c,a}$ and $\eta_{c,c}$ respectively are concentration overpotentials at anode and cathode. Parameters i_{la} and i_{lc} respectively are anode and cathode limiting current densities. Also, $D_{a,eff}$ and $D_{c,eff}$ are effective gaseous diffusivities through the anode and cathode respectively, while δ_a and δ_c are thicknesses of the anode and cathode respectively.

Sources: [38,39].

steady state conditions and neglecting and potential energy values can be written as:

$$\dot{Q} - \dot{W} + \sum_{in} \dot{N}_i \bar{h}_i - \sum_{out} \dot{N}_o \bar{h}_o = 0 \quad (37)$$

where $\dot{N}$ and $\bar{h}$ respectively denote mole flow rate and molar enthalpy, and $\dot{Q}$ and $\dot{W}$ respectively denote heat transfer rate and work rate. The isentropic efficiencies for the air/methane compressors and water pump are all assumed to be 85 % [32]. The pressure drop in the afterburner and heat exchangers are assumed to be 3 % and 2 % respectively of the corresponding input pressures [31].

The ambient temperature and pressure vary according to altitude (Z), with both decreasing with increasing altitude. The ambient temperature and ambient pressure can be determined as follows:

$$T_a = 288.15 + L_a Z \quad (38)$$

$$P_a = 101.325 \left(\frac{288.15}{T_a} \right)^{\frac{g}{L_a R_a}} \quad (39)$$

where R_a , g and L_a are the characteristic gas constant for air, gravitational acceleration and the base temperature lapse rate per kilometer of geopotential altitude ($L_a = 6.5$ K/km) [7]. The inlet air condition to the system is calculated as:

$$P_{air,in} = P_a \left[1 + \frac{(\gamma_a - 1)}{2} M^2 \right]^{(\gamma_a - 1)^{-1}} \quad (40)$$

$$T_{air,in} = T_a \left[1 + \frac{(\gamma_a - 1)}{2} M^2 \right] \quad (41)$$

where γ_a and M respectively are the specific heat ratio of air (1.4) and Mach number. The Mach number for an altitude of 10 km is 0.83 [7].

It is assumed that 40 % of the combusted flue gas leaving the afterburner is transferred to the gas turbine for electrical power generation and that the remaining portion (60 %) is conveyed to the heat exchangers for preheating of SOFC reactants. The gas turbine outlet temperature is determined as follows:

$$T_{20} = T_{16} \left[1 - \eta_{isen,T} \left[1 - \left(\frac{P_{16}}{P_{20}} \right) \right] \right] \quad (42)$$

where $\eta_{isen,T}$ is the isentropic efficiency of the gas turbine, considered here to be 88 %. The net electrical power output of the SOFC and gas turbine can be obtained as the difference between the total electrical power produced by the gas turbine and SOFC and the total required power for pumps and compressors (and any other devices requiring electricity). That is,

$$\dot{W}_{net} = \dot{W}_{SOFC} + \dot{W}_{GT} - \sum_{compressors} \dot{W}_i + \sum_{pumps} \dot{W}_j \quad (43)$$

The net fuel-to-electricity conversion efficiency compares net power produced by the SOFC-battery-gas turbine propulsion system to the input fuel consumption, as follows:

$$\eta_{SOFC-GT} = \frac{\dot{W}_{net}}{\dot{n}_{CH_4,in} \overline{LHV}_{CH_4}} \quad (44)$$

where $\overline{LHV}_{CH_4}$ is low heating value of methane and $\dot{n}_{CH_4,in}$ is the molar flow rate of methane to the hybrid propulsion system.

4. Results and discussion

The results of the modeling and analysis of the SOFC-battery hybridized gas turbine propulsion system for electric aircraft are presented and discussed in this section. The aircraft is assumed to be in cruising condition at an altitude of 10 km, where the ambient conditions are 254 K and 36.45 kPa. More specifically, the effects are assessed and reported

of varying the main system parameters, like SOFC temperature and pressure and current density, on the net electrical power produced and the energy efficiency for the hybrid SOFC-battery-gas turbine. For the case of battery pack used in a hybrid electric aircraft, resolving full three-dimensional model of hundreds of lithium ion battery cell is computationally costly. So, for simplicity, the simulation results of single cell of lithium ion battery is presented here.

The insensitivity of the battery model to the mesh is ensured by performing a mesh independence study. In this study, the sensitivity of the maximum temperature of the lithium ion battery at the end of the charging duration on the mesh is examined. Fig. 5 illustrates that the maximum temperature of the battery cell changes by less than 0.7 % as the number of mesh elements reaches 15079. COMSOL Multiphysics software is used to mesh and simulate the lithium ion battery with an initial temperature of 298 K and an initial state of charge of 20 %. Also, the charge and discharge durations are taken to be 600 s each.

The lithium ion battery electrochemical model is validated through the comparison between the variations of the battery cell voltage against the attainable capacity for the cell at different discharge rates with the experimental results presented in Doyle et al. [28]. The comparison between the simulation results of the present study and those experimental results from Doyle et al. [28] are shown in Fig. 6. Various discharge rates are presented varying from a low discharge rate of 0.1C to a discharge rate of 2C. As shown in Fig. 6 the developed model in this study accurately predicts the electrochemical behavior of the lithium ion battery. In addition, the methane-fueled solid oxide fuel cell system developed in the present study is validated through experimental data reported elsewhere. In particular, Table 6 compares the changes of cell voltage and power density with current density in the present study with the corresponding experimental results of Tao et al. [33]. It is seen that the simulation results in this study exhibit good agreement with the experimental data presented by Tao et al., [33], with an error of less than 5.3 %.

In addition, the results of a thermal study of the lithium ion battery are presented. The thermal model of the lithium ion battery is simulated in Comsol multi-physics software. The average values for the temperature and heat generation rate in the lithium ion battery are transferred to Engineering Equation Solver (EES), where the SOFC stack, afterburner, gas turbine and compressors are modeled. The simulation results presented in this section for the lithium ion battery are for a discharge rate of 16 A, an initial temperature of 298 K and an initial state of charge of 20 %. The charge and discharge durations are taken to be 600 s each.

Fig. 7 shows a single charge-discharge cycle for the lithium ion battery. The charge and discharge durations are 600 s each. A positive current for the battery is associated with charging and a negative sign with discharging. Fig. 8 compares the battery open circuit potential and cell potential, as they vary with time. It is seen that the battery open

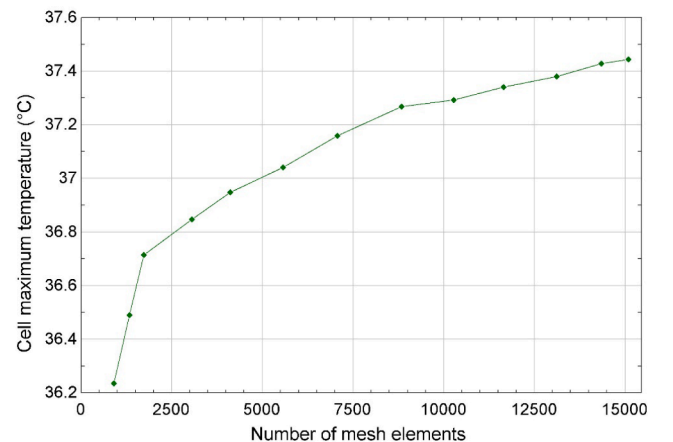


Fig. 5. Mesh independence study for simulation of lithium ion battery.

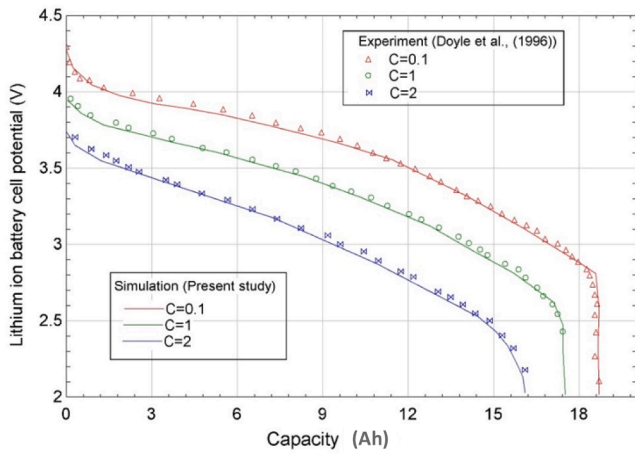


Fig. 6. Comparison of the variation of the lithium ion battery potential with the capacity of the battery at various discharge rates for the model developed in the present study and the experimental results of Doyle et al. [28].

Table 6

Comparison of methane-fueled SOFC model developed in present study with the experimental results of Tao et al. [33].

Current density (A/ m ²)	Cell voltage (V)		Power density (W/m ²)	
	Present study	Tao et al. [33]	Present study	Tao et al. [33]
0.2	0.73	0.76	0.146	0.152
0.4	0.60	0.62	0.24	0.248
0.6	0.51	0.52	0.306	0.312

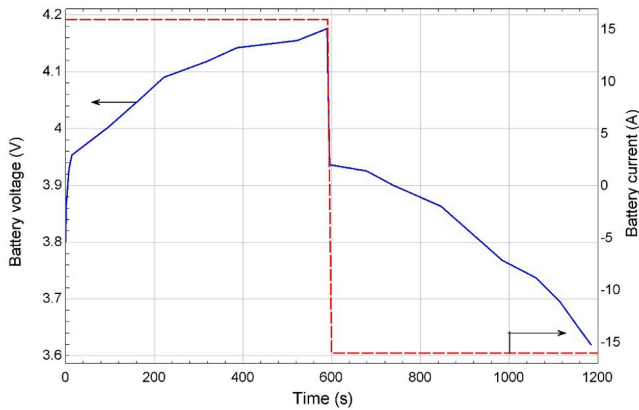


Fig. 7. Variation with time of lithium ion battery cell potential and the load cycle current.

circuit potential is higher than the cell potential during discharge and is lower during charge because during the charge and discharge operations of the battery, the cell potential experiences ohmic, surface and concentration overpotentials.

Fig. 9 shows the variation of battery maximum temperature during discharging and charging periods for several C-rates. It is seen that the battery temperature increases consistently over the operation of battery. In addition, higher temperatures are observed at higher C-rates. For instance, at a time of 500 s, the temperature of the lithium ion battery increases from 27 °C to 44 °C as the C-rate increases from 1 to 4. This graph is particularly useful as it provides the maximum allowable temperature of the battery that can be tolerated to prevent melting of solid parts of the electrodes or thermal decomposition of the polymer in the lithium ion battery.

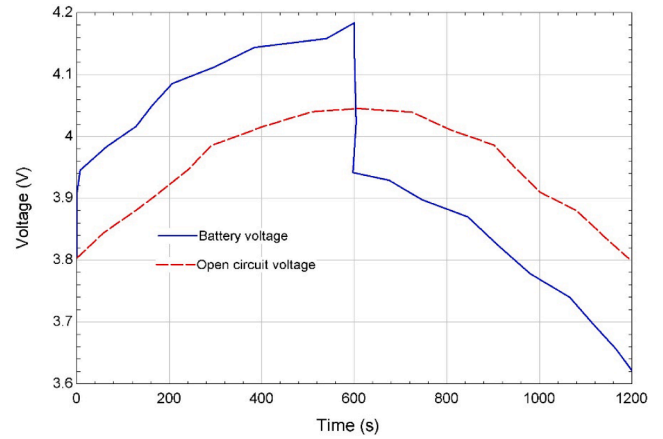


Fig. 8. Cell potential and open circuit potential variations with time during operation of the lithium ion battery cell.

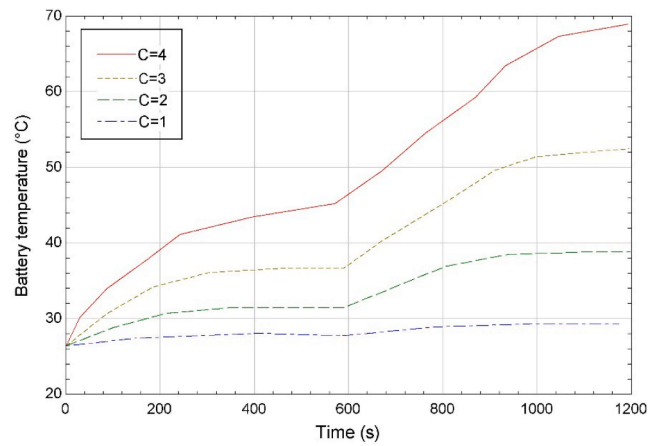


Fig. 9. Variation of battery maximum temperature with charging and discharging time for several C-rates.

Fig. 10 shows the variations with time of the heat generation rate per volume (i.e., volumetric heat generation rate) for the lithium ion battery during charging and discharging periods, for several C-rates. According to Fig. 10, during charging, the volumetric heat generation rate decreases because the increase in the battery temperature (Fig. 9) reduces the overpotentials, which constitute the primary ohmic loss. Then near the end of charging, the volumetric heat generation rate increases as the

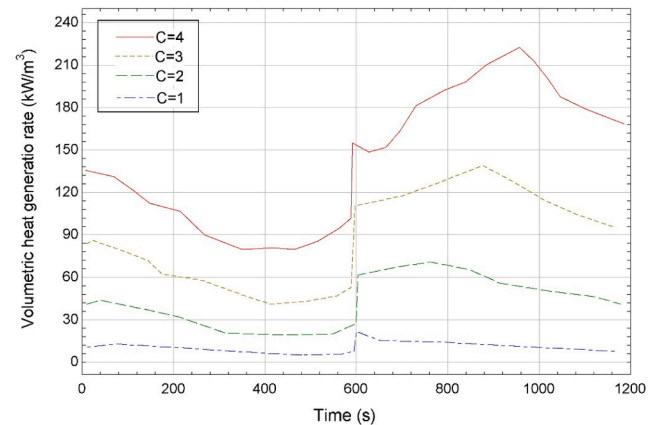


Fig. 10. Variation of battery volumetric heat generation rate with charging and discharging time for several C-rates.

further increase in battery temperature (Fig. 9) leads to an increase in the overpotentials. Also, during discharging, the volumetric heat generation rate in the battery increases due to the increase in the overpotentials (mainly ohmic overpotential) and then this heat rate decreases due to decrease in the overpotentials (mainly concentration overpotential). Note that the battery temperature increases consistently during both charge and discharge (Fig. 9).

Fig. 11 illustrates the temperature distribution through a single lithium ion battery cell selected times. It is seen that the temperature of various parts of battery does not change significantly. For instance, at the end of discharge (1119 s), the difference between battery maximum and minimum temperatures is 4 °C. This demonstrates the validity of the assumption of uniform temperature throughout the lithium ion battery cell.

Fig. 12 shows the variations with the current density in the SOFC of the electrical power produced by the SOFC and gas turbine as well as the power consumed by the air compressor, fuel compressor and water pumps. It is seen that the slope of the increase is lower for the SOFC power production than the gas turbine and air compressor. The reason is that increasing the current density raises the overpotentials, primary activation and ohmic losses in the SOFC, reducing the fuel cell potential. But the decline in the cell potential is smaller than the rise in the current density, which results in an increase in the electrical power output by the SOFC. It is also seen that the increase in the current density of the SOFC leads to an increase in all of the electrical power producer/consumer components. Higher current densities result in higher flow rates of reactants (air, methane and water) and consequently more power is required to pressurize those reactants to the SOFC pressure. The electrical power consumed for the fuel compressor and water pump is lower than that for the air compressor, as the input rate is much higher for air than for methane and water.

Fig. 13 shows the variations with SOFC current density of total net electrical power yielded by the hybrid SOFC-battery-gas turbine system and its energy efficiency in the electric aircraft. It is observed that the net electrical power produced by the hybrid system is a maximum (6.7 MW) at a current density of 10,000 A/m². However, the energy efficiency of the hybrid system decreases continually with increasing current density. This is mainly due to the fact that the increase in the input flow rate of methane at higher current densities is greater than the increase in the net electrical power output from the hybrid system. Therefore, it is suggested that the current density of SOFC not exceed 10,000 A/m².

Fig. 14 shows the effect of pressure ratio on the net electrical power produced by and energy efficiency of the hybrid system. The pressure

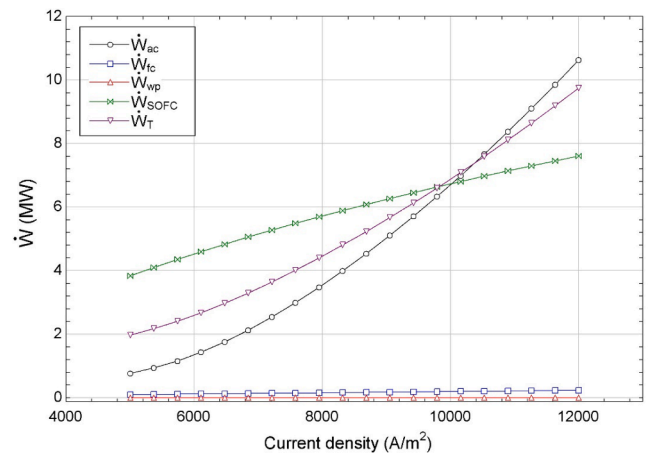


Fig. 12. Variations with the current density in the SOFC of the electrical power production/consumption components for the hybrid SOFC-battery-gas turbine system.

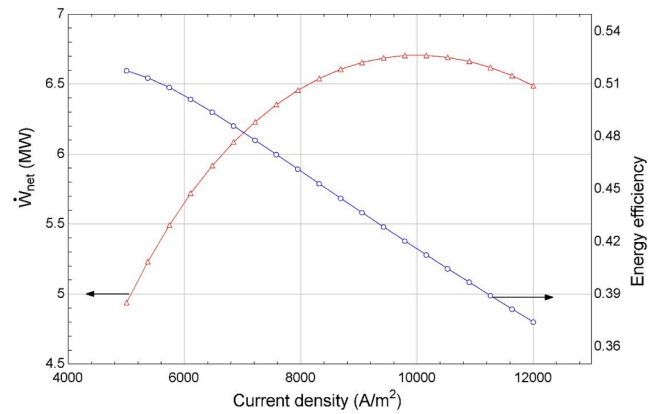


Fig. 13. Variations of net electrical power produced by and energy efficiency of the hybrid SOFC-battery-gas turbine system with SOFC current density.

ratio indicates the ratio of output to input pressure of the SOFC reactants, and is applicable to the compressors and pump. As stated earlier, the SOFC reactants of water, methane and air are pressurized to

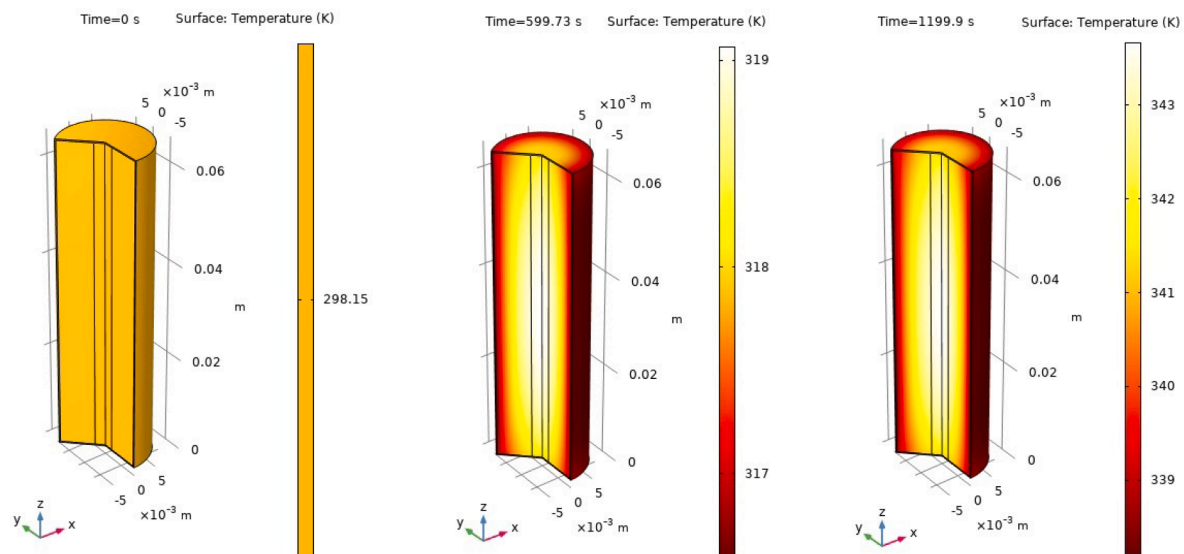


Fig. 11. Temperature distribution in the lithium ion battery cell at beginning of charging (left), end of charging (center) and end of discharging (right).

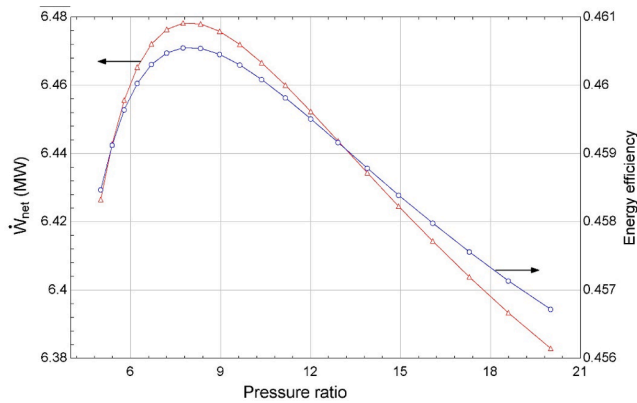


Fig. 14. Variation with pressure ratio of SOFC reactants of net electrical power produced and energy efficiency for the hybrid SOFC-battery-gas turbine system.

the SOFC pressure before entering the internal reformer. A higher pressurization ratio results in higher electrical power production by the SOFC and gas turbine, as well as higher power consumption by the air/fuel compressors and water pump. As a consequence, the net power produced from the system reaches maximum at a specific pressurization ratio. The energy efficiency of the hybrid system is also a maximum at that pressurization ratio. From Fig. 14 it is seen that the maximum net power output and energy efficiency are achieved at pressurization ratio of approximately 7.5.

Fig. 15 shows the variation of electrical power produced or consumed by the hybrid system components, with the temperature of SOFC. It can be observed that the electrical power produced by the SOFC increases slightly as the SOFC temperature reaches 700 °C, and then starts to decrease above this temperature. The reason is that the voltage (or potential) losses in the SOFC – including ohmic, activation and concentration overpotentials – decline as the temperature rises to 700 °C, and then rise as SOFC temperature further increases. In addition, the increase in the voltage losses with increasing SOFC temperature up to 700 °C results in an increase in the heat loss in the SOFC in which the energy of SOFC products declines. This leads to a reduction in the electrical power production by the gas turbine as can be observed as the SOFC temperature rises to 700 °C. At that temperature, the electrical power produced at the gas turbine reaches its minimum value.

Fig. 16 shows the variations in net electrical power produced by and energy efficiency of the hybrid system, with SOFC temperature. The net power production and the energy efficiency curves for the system exhibit very similar profile variations with SOFC temperature. Both

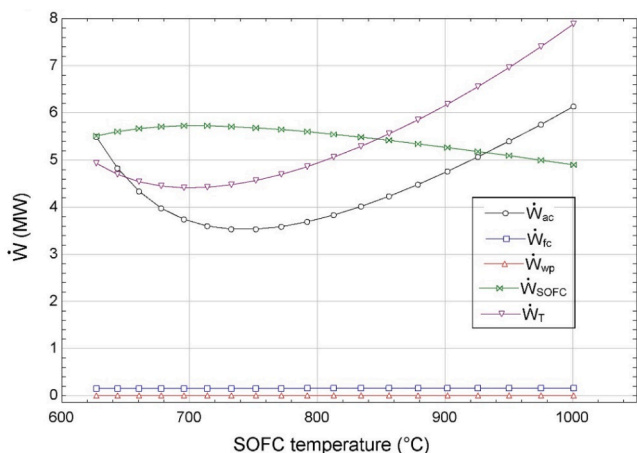


Fig. 15. Variations with SOFC temperature of electrical power productions and consumptions for components of the hybrid SOFC-battery-gas turbine system.

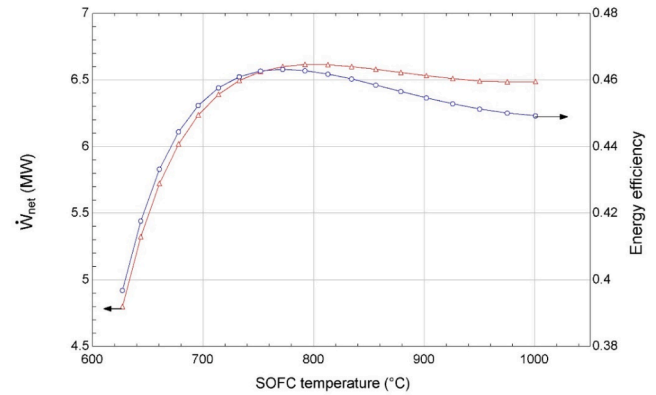


Fig. 16. Variations with SOFC temperature of net electrical power production and energy efficiency for the hybrid SOFC-battery-gas turbine system.

parameters reach their maximum values at an SOFC temperature of approximately 775 °C. So, it is suggested that the SOFC operates at this temperature. The net electrical power produced by and energy efficiency of the hybrid system increase respectively from 4.75 MW to 6.6 MW and 39.5 % to 46.2 % as the SOFC temperature increases from 627 °C to 775 °C.

5. Conclusions

This paper illustrates the feasibility of a novel design of a propulsion system for an electric aircraft in which a SOFC-battery system is hybridized with a gas turbine. Lithium ion batteries are used to supply electrical power during operating conditions having high power loads like climb and takeoff. The system is designed accounting for the cooling process required for the lithium ion battery pack to increase the lifetime and active operation of batteries in electric aircraft. The effect of C-rate on the temperature and heat generation flux is described. The average values for the temperature and heat generation flux are used in the simulation of hybrid electric propulsion system. Through a parametric analysis of the system and its components, the effects are described of varying several operating conditions, such as SOFC temperature and pressure and current density, on the net electrical power produced by the system and its energy efficiency. Optimum values at which the maximum net power production and energy efficiency are achieved are identified. At 1C, water is capable of removing 15,000 W/m³ per volume of lithium ion battery to maintain the temperature at about 28 °C. In addition, the net electrical power produced and energy efficiency of the hybrid propulsion system reach their maximum values (i.e., 6.6 × 10⁶ W and 46.5 %, respectively) at a SOFC temperature of approximately 775 °C. The net electrical power produced by the hybrid system is a maximum (6.7 MW) at a current density of 10,000 A/m²; however, the energy efficiency of the hybrid system decreases continually with increasing current density. So, it is suggested that the SOFC current density not exceed 10,000 A/m².

Further research on optimum battery energy storage capacity, airframe design and dynamics, use of pressurized SOFC, system packaging and aircraft subsystems is required to support development and commercial implementation of hybrid electric aircrafts and to ensure safety and reliability of them. Also, designing a light-weight battery thermal management system could improve the achievable payload. Overall, the analysis of the proposed hybrid propulsion system for electric aircraft suggests that SOFC-battery-gas turbine hybrids may have a role in future commercial aircraft.

CRedit authorship contribution statement

Aida Farsi: Conceptualization, Methodology, Software, Writing – original draft. **Marc A. Rosen:** Conceptualization, Writing – review &

editing, Supervision, Project administration.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Aida Farsi reports financial support was provided by Natural Sciences and Engineering Research Council of Canada. The authors have no conflicts of interest to declare that are relevant to the content of this article.

Data availability

The data that has been used is confidential.

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